

练习 4.7.1: 为练习 4.2.1 的文法 $S \rightarrow SS + | SS * | a$ 构造

1) 规范 LR 项集族。

2) LALR 项集族。

1) 增广文法: $S' \rightarrow S$
 $S \rightarrow SS +$
 $S \rightarrow SS *$
 $S \rightarrow a$

构造项集族

$$\begin{aligned} \textcircled{1} I_0 &= \text{CLOSURE}(\{[S' \rightarrow \cdot S, \$]\}) \quad [A \rightarrow \alpha \cdot B\beta, a] \\ &= \{[S' \rightarrow \cdot S, \$], \\ &\quad [S \rightarrow \cdot SS+, \$], [S \rightarrow \cdot SS*, \$], [S \rightarrow \cdot a, \$], \\ &\quad [S \rightarrow \cdot SS+, a], [S \rightarrow \cdot SS*, a], [S \rightarrow \cdot a, a]\} \\ &= \{[S' \rightarrow \cdot S, \$], [S \rightarrow \cdot SS+, \$/a], \\ &\quad [S \rightarrow \cdot SS*, \$/a], [S \rightarrow \cdot a, \$/a]\} \end{aligned}$$

$\textcircled{2} \text{GOTO}(I_0, S)$

$$\begin{aligned} &= \{[S' \rightarrow S \cdot, \$], [S \rightarrow S \cdot S+, \$/a], [S \rightarrow S \cdot S*, \$/a], \\ &\quad [S \rightarrow \cdot SS+, +], [S \rightarrow \cdot SS*, +], [S \rightarrow \cdot a, +], \\ &\quad [S \rightarrow \cdot SS+, *], [S \rightarrow \cdot SS*, *], [S \rightarrow \cdot a, *], \\ &\quad [S \rightarrow \cdot SS+, a], [S \rightarrow \cdot SS*, a], [S \rightarrow \cdot a, a]\} \\ &= \{[S' \rightarrow S \cdot, \$], [S \rightarrow S \cdot S+, \$/a], [S \rightarrow S \cdot S*, \$/a], \\ &\quad [S \rightarrow \cdot SS+, +/*/a], [S \rightarrow \cdot SS*, +/*/a], [S \rightarrow \cdot a, +/*/a]\} \\ &= \textcircled{1} \end{aligned}$$

$$\text{GOTO}(I_0, a) = \{[S \rightarrow a \cdot, \$/a]\} = \textcircled{2} \checkmark$$

$$\text{GOTO}(I_0, +) = \text{GOTO}(I_0, *) = \emptyset$$

$\textcircled{3} \text{GOTO}(I_1, S)$

$$\begin{aligned} &= \{[S \rightarrow SS \cdot, +, \$/a], [S \rightarrow SS \cdot, *, \$/a], \\ &\quad [S \rightarrow S \cdot S+, +/*/a], [S \rightarrow S \cdot S*, +/*/a], \\ &\quad [S \rightarrow \cdot SS+, +/*], [S \rightarrow \cdot SS*, +/*], \\ &\quad [S \rightarrow \cdot a, +/*], \\ &\quad [S \rightarrow \cdot SS+, a], [S \rightarrow \cdot SS*, a], \\ &\quad [S \rightarrow \cdot a, a]\} \\ &= \{[S \rightarrow SS \cdot, +, \$/a], [S \rightarrow SS \cdot, *, \$/a], \\ &\quad [S \rightarrow S \cdot S+, +/*/a], [S \rightarrow S \cdot S*, +/*/a], \\ &\quad [S \rightarrow \cdot SS+, +/*/a], [S \rightarrow \cdot SS*, +/*/a], \\ &\quad [S \rightarrow \cdot a, +/*/a]\} = \textcircled{3} \end{aligned}$$

$$\text{GOTO}(I_1, a) = \{[S \rightarrow a \cdot, +/*/a]\} = \textcircled{4} \checkmark$$

$$\text{GOTO}(I_1, +) = \text{GOTO}(I_1, *) = \emptyset$$

$\textcircled{4} \text{GOTO}(I_2, S)$

$$\begin{aligned} &= \{[S \rightarrow SS \cdot, +, +/*/a], [S \rightarrow SS \cdot, *, +/*/a], \\ &\quad [S \rightarrow S \cdot S+, +/*/a], [S \rightarrow S \cdot S*, +/*/a], \\ &\quad [S \rightarrow \cdot SS+, +/*], [S \rightarrow \cdot SS*, +/*], \\ &\quad [S \rightarrow \cdot a, +/*], \\ &\quad [S \rightarrow \cdot SS+, a], [S \rightarrow \cdot SS*, a], \\ &\quad [S \rightarrow \cdot a, a]\} \\ &= \{[S \rightarrow SS \cdot, +, +/*/a], [S \rightarrow SS \cdot, *, +/*/a], \\ &\quad [S \rightarrow S \cdot S+, +/*/a], [S \rightarrow S \cdot S*, +/*/a], \\ &\quad [S \rightarrow \cdot SS+, +/*/a], [S \rightarrow \cdot SS*, +/*/a], \\ &\quad [S \rightarrow \cdot a, +/*/a]\} = \textcircled{5} \end{aligned}$$

$$\text{GOTO}(I_2, a) = \{[S \rightarrow a \cdot, +/*/a]\} = I_4$$

$$\text{GOTO}(I_2, +) = \{[S \rightarrow SS+, \$/a]\} = \textcircled{6} \checkmark$$

$$\text{GOTO}(I_2, *) = \{[S \rightarrow SS*, \$/a]\} = \textcircled{7} \checkmark$$

$\textcircled{5} \text{GOTO}(I_5, S)$

$$\begin{aligned} &= \{[S \rightarrow SS \cdot, +, +/*/a], [S \rightarrow SS \cdot, *, +/*/a], \\ &\quad [S \rightarrow S \cdot S+, +/*/a], [S \rightarrow S \cdot S*, +/*/a], \\ &\quad [S \rightarrow \cdot SS+, +/*/a], [S \rightarrow \cdot SS*, +/*/a], \\ &\quad [S \rightarrow \cdot a, +/*/a]\} = I_5 \end{aligned}$$

$$\text{GOTO}(I_5, a) = I_4$$

$$\text{GOTO}(I_5, +) = \{[S \rightarrow SS+, +/*/a]\} = \textcircled{8} \checkmark$$

$$\text{GOTO}(I_5, *) = \{[S \rightarrow SS*, +/*/a]\} = \textcircled{9} \checkmark$$

2) 具有不同核心的 LR 项:

$$\{\{I_0\}, \{I_1\}, \{I_2, I_4\}, \{I_3, I_5\}, \{I_6, I_8\}, \{I_7, I_9\}\}$$

合并同核心项:

$$I_0, I_1,$$

$$I_{24} = \{[S \rightarrow a \cdot, +/*/a/\$]\}$$

$$\begin{aligned} I_{35} = \{[S \rightarrow SS \cdot, +, +/*/a/\$], [S \rightarrow SS \cdot, *, +/*/a/\$], \\ [S \rightarrow S \cdot S+, +/*/a], [S \rightarrow S \cdot S*, +/*/a], \\ [S \rightarrow \cdot SS+, +/*/a], [S \rightarrow \cdot SS*, +/*/a], \\ [S \rightarrow \cdot a, +/*/a]\} \end{aligned}$$

$$I_{68} = \{[S \rightarrow SS+, +/*/a/\$]\}$$

$$I_{79} = \{[S \rightarrow SS*, +/*/a/\$]\}$$

和 SLR 项数相同。

！练习 4.7.4：说明下面的文法

$$\begin{aligned} S &\rightarrow A a \mid b A c \mid d c \mid b d a \\ A &\rightarrow d \end{aligned}$$

是 LALR(1) 的，但不是 SLR(1) 的。

证：a. SLR(1)

构造 LR(0) 规范项集族：

$$I_0 = \text{CLOSURE}(\{[S' \rightarrow S]\})$$

$$= \{[S' \rightarrow S], [S \rightarrow \cdot A a], [S \rightarrow \cdot b A c], [S \rightarrow \cdot d c], [S \rightarrow \cdot b d a], [A \rightarrow \cdot d]\} \quad \textcircled{0} b d \textcircled{0} d$$

注意到存在且仅存在两个移入-规约冲突。

$$\textcircled{1} \text{GOTO}(I_0, b) = \{[S \rightarrow b \cdot A c], [S \rightarrow b \cdot d a], [A \rightarrow \cdot d]\} = I_1$$

$$\text{GOTO}(I_1, d) = \{[S \rightarrow b d \cdot a], [A \rightarrow d \cdot]\}$$

$$\textcircled{2} \text{GOTO}(I_0, d) = \{[S \rightarrow d \cdot c], [A \rightarrow d \cdot]\}$$

$$\text{FOLLOW}(S) = \{\$, \}$$

$$\text{FOLLOW}(A) = \{a, c\}$$

对于 $\textcircled{1}$ ，在 I_1 状态下输入 a 时，由于 $a \in \text{FOLLOW}(A)$ ，因此既可移入也可规约。

$\textcircled{2}$ 同理，因为 $c \in \text{FOLLOW}(A)$ 。

故不是 SLR(1) 文法。

b. LALR(1)

在 LALR(1) 中

$$\text{GOTO}(I_0, d) = \{[S \rightarrow d \cdot c, \$], [A \rightarrow d \cdot, c]\}$$

$$\text{GOTO}(I_1, d) = \{[S \rightarrow b d \cdot a, \$], [A \rightarrow d \cdot, a]\}$$

不会发生冲突，故是 LALR 文法。

□

！练习 4.7.5：说明下面的文法

$$\begin{aligned} S &\rightarrow A a \mid \underline{b A c} \mid B c \mid \underline{b B a} \\ A &\rightarrow d \\ B &\rightarrow d \end{aligned}$$

是 LR(1) 的，但不是 LALR(1) 的。

证：只需证合并同核心项会导致冲突即可。

通过构造 LR(1) 项可说明该文法是 LR(1) 的。

下面考虑合并导致冲突的情况。

$$I_0 = \text{CLOSURE}(\{[S' \rightarrow S, \$]\})$$

$$= \{[S' \rightarrow S, \$], [S \rightarrow \cdot A a, \$], [S \rightarrow \cdot b A c, \$], [S \rightarrow \cdot B c, \$], [S \rightarrow \cdot b B a, \$], [A \rightarrow \cdot d, a], [B \rightarrow \cdot d, c]\}$$

$$I_1 = \text{GOTO}(I_0, d) = \{[A \rightarrow d \cdot, a], [B \rightarrow d \cdot, c]\}$$

$$I_2 = \text{GOTO}(I_0, b) = \{[S \rightarrow b \cdot A c, \$], [S \rightarrow b \cdot B a, \$], [A \rightarrow \cdot d, c], [B \rightarrow \cdot d, a]\}$$

$$I_3 = \text{GOTO}(I_2, d) = \{[A \rightarrow d \cdot, c], [B \rightarrow d \cdot, a]\}$$

合并 I_1, I_3 ：

$$I_{13} = \{[A \rightarrow d \cdot, a/c], [B \rightarrow d \cdot, a/c]\}$$

会发生冲突。

故不是 LALR(1) 文法。

□